



Development and validation of a Game-Induced Happiness Scale (GIHS) based on the actual conditions of university students

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ABSTRACT

Online games are growing rapidly, but their impact on players' well-being has not been sufficiently studied due to the lack of specialized measurement tools. This study aimed to address this gap by developing a "Game-Induced Happiness Scale (GIHS)" and verifying its validity. After conducting a preliminary survey with 143 participants to develop the items, we administered a questionnaire survey to 500 college students who regularly play digital games. Exploratory factor analysis (EFA) confirmed the existence of five factors: "Factor I: Social Interaction and Communication," "Factor II: Strategic Gameplay and Skill Enhancement," "Factor III: Character Appeal and Personality," "Factor IV: Long-Term Immersion and Engagement," and "Factor V: In-Game Economy and Rewards." These results demonstrated high reliability and validity. These findings provide important insights for game developers to enhance user experience and support practitioners in managing potential negative game effects.

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1. Introduction

1.1. The purpose of this research

The digital gaming industry is rapidly expanding within the global entertainment market, playing a significant role in various fields such as education, psychological health, and the promotion of social connections. Specifically, the impact of games on users' happiness, encompassing both positive and negative aspects, has garnered considerable attention from researchers. In this study, we focused on developing a novel scale to measure game-induced happiness, "Game-Induced Happiness Scale (GIHS)", and to further evaluate the scale's reliability and validity. By doing so, we aimed to elucidate the structure of happiness fostered by digital games and theoretically explore how gaming experiences affect psychological satisfaction and well-being. Furthermore, this study examined the potential applications of these findings in education, the promotion of psychological health, and the prevention of gaming addiction. Ultimately, we aimed to reassess the social value of gaming and provide foundational knowledge for building a healthy gaming culture. In this paper, happiness, well-being, and subjective well-being (SWB) are used interchangeably.

2. Research background

2.1. Scale and reach of digital games

Digital games today constitute the world's largest entertainment sector. The global player base stood at 3.42 billion in 2024 and is projected to reach 3.76 billion by 2027, a roughly 10% increase from 2024, and 2024 revenues amounted to \$187.7 billion [1].

Growth accelerators include cloud distribution, free-to-play business models, and the professionalization of eSports, each of which widens demographic access and embeds games in everyday life.

2.2. Documented effects of gameplay

Research portrays a double-edged sword of outcomes. Positive facets involve cognitive gains (e.g. faster visual attention [2]), improved probabilistic learning (a facet of problem-solving) [3], and social connectedness fostered by cooperative play [4]. Educational applications – from Minecraft to physics simulators – demonstrate enhanced motivation and learning [5]. Conversely, negative facets include excessive use leading to sleep disruption, impaired functioning,

and the WHO-recognized "gaming disorder" affecting a few percent of players [6].

Darvesh et al. reported that, based on a review of a wide range of prior studies on the prevalence of gaming disorder, the estimated prevalence of gaming disorder in the general population varies significantly across studies, and that differences exist depending on the target population (general vs. clinical). [7].

2.3. Why quantify happiness in games in this study

According to the UK Digital, Culture, Media, and Sport Committee [8], recent policy discussions on gaming have shifted focus from the link between violent content and aggression to the relationship between game characteristics, playtime, and well-being. Understanding game-induced happiness may help mitigate risks such as gaming disorder and social isolation while promoting balanced play.

Furthermore, considering the dual nature of games, happiness becomes an important criterion for determining the net benefits of playing. A game-induced happiness scale will

1. inform balanced game design that amplifies well-being while minimizing harm,
2. guide policymakers beyond simplistic "time-spent" regulations, and
3. equip educators and therapists to select or adapt titles that genuinely enhance learner or client flourishing.

Accordingly, the present study develops and validates the Game-Induced Happiness Scale (GIHS) to provide an evidence-based yardstick for these endeavours.

2.4. Literature review of happiness

In the previous section, we emphasized both the positive and negative aspects of games, and emphasized that it is important to address these factors to properly support users' game-related behavior. Through this research, we aimed to provide a foundation for game design that fosters happiness and allows for future exploration of its potential applications in education and psychological settings. Happiness in the context of gaming is often understood as the positive emotional state derived from gameplay. The following sections discuss the definition of happiness and its theoretical background, along with its relationship to gaming.

However, the present scale was developed inductively; no a-priori theoretical model (e.g., PERMA, SDT, Flow) constrained item generation. These discussions remain useful for post-hoc interpretation. (For

how the emergent factors may align with real-life well-being constructs (see Section 5.2))

2.4.1. Definition and theoretical background of happiness

Happiness plays a significant role not only in gaming but also in everyday life [9]. Extensive research has explored the concept of happiness, resulting in various definitions. Diener [9] describes happiness as “the totality of subjective pleasure and satisfaction”, “the degree to which an individual feels content with their life”, and “the balance between positive and negative emotions”. This definition exemplifies a subjective approach to understanding happiness. At a more granular level, contemporary psychology distinguishes between hedonic happiness (momentary pleasure) and eudaimonic well-being (meaning, self-realization) [10].

SDT [11] posits that autonomy (volition), competence (effectance), and relatedness (social connection) are innate psychological nutrients; satisfying them nurtures both hedonic pleasure and eudaimonic growth. Furthermore, numerous studies have linked behavioral economics and well-being (e.g., [12]), and SDT is expected to bridge conventional utilitarian (economic) approaches with research on emotional well-being.

Seligman [13] proposed the PERMA model, which conceptualizes well-being through five clusters-Positive Emotion, Engagement, Relationships, Meaning, and Accomplishment/Achievement. Positive Emotion refers to joy or comfort; Engagement captures deep absorption akin to “flow” [14]; Relationships cover supportive social ties; Meaning involves belonging to something larger than oneself; and Accomplishment denotes progress toward valued goals. One widely used method for measuring happiness, therefore, is the “Subjective Well-Being Scale (SWLS)” [15], which targets the cognitive component of well-being and complements PERMA’s multi-facet approach.

Csikszentmihalyi’s Flow [14] describes an optimal state of consciousness characterized by intense focus, distortion of time, and intrinsic reward, occurring when perceived skill matches challenge. Flow has been framed as a mechanism connecting momentary engagement with lasting fulfillment, complementing PERMA’s Engagement.

Numerous studies in behavioral economics emphasize the importance of subjective happiness/well-being, focusing on its impact on decision-making and quality of life [12]. Happiness has also been linked to physical health: Tönis et al. [16] showed that positive psychological states improved behavior and quality of life among cardiovascular patients. Happiness is further associated with psychological resilience and stress management [17,18]. According to Fredrickson’s “Broaden-and-Build Theory”, [17] positive emotions expand cognitive and social resources, enhancing adaptability in challenging situations. These findings highlight the multifaceted benefits of happiness at individual and societal levels.

As described above, there are many previous studies on happiness, and these studies are expected to show a certain correlation with happiness in games.

2.4.2. The relationship between existing theories of happiness and happiness in games

Happiness in everyday life is believed to be directly or indirectly related to happiness in games. Here, we discuss content that is assumed to be directly or indirectly related. However, considering the diversification of current game styles, it is expected that there are feelings of happiness that were not captured in previous studies. Therefore, the purpose of this study is to develop based on the actual situations of the survey participants. Thus, rather than setting hypotheses, we present content that is assumed to be related.

Deci’s SDT provides a significant lens: autonomy (freedom of choice), competence (skill mastery), and relatedness (social bonding) are often fulfilled during gameplay [19]. Open-world titles grant autonomy; escalating difficulty curves cultivate competence; cooperative

raids in MMORPGs strengthen relatedness [4]. Empirical analytic evidence indicates that SDT need satisfaction mediates the link between play and post-game positive affect [20].

Each pillar of PERMA can similarly be found in digital games. Casual titles are expected to heighten Positive Emotions; tightly regulated esports foster Engagement; guild systems nurture Relationships; story-driven RPGs provide Meaning; and Achievement, trophy systems deliver a sense of Accomplishment.

Johannes et al. [21] reported increased daily life satisfaction after brief sessions of Animal Crossing [22]. While their study did not frame the findings in terms of happiness theory, the observed affective improvement is consistent with PERMA-P (Positive Emotion), the first pillar of Seligman’s PERMA model representing joy and pleasure [13].

Flow theory predicts that games offering immediate feedback and calibrated challenge yield deeper satisfaction [14]. Experimental work shows that flow-intensity ratings correlate with post-session vitality and reduced stress [23]. Iconic examples include “perfect-dodge” mechanics in Bayonetta [24] (where players evade enemy attacks with precise timing to trigger a slow-motion counter window) or rhythm-matched levels in Beat Saber [25] (where synchronized body movements to music generate flow-like immersion).

Factors such as social interaction, competence development, autonomy support, and in-game immersion have been shown to contribute to positive player experience and well-being [20,26]. Granic et al. [19] highlight that for most individuals games are a positive leisure pursuit, yet excessive use can disrupt circadian rhythms and diminish well-being.

Digital games also feature a variety of rewards, such as in-game currency, experience points, items, badges, and recognition from other players. Each of these plays a role in motivating player behavior and increasing engagement [20,27]. Many of these rewards can be considered examples of “extrinsic motivation”, which is motivation that comes from external sources to encourage specific actions or continued play [5,26]. On the other hand, the sense of satisfaction and enjoyment derived from the game experience itself, such as the sense of accomplishment when overcoming challenges, improvements in one’s skills, and immersion in the game world, are classified as “intrinsic motivation”. Such intrinsic motivation is deeply related to psychological needs such as autonomy, competence, and relatedness, and has been shown in many studies to strongly contribute to long-term satisfaction and happiness [20,26]. In addition to established models such as SDT and PERMA, recent work has introduced theory-based instruments for assessing need satisfaction during gaming. For example, the Basic Needs in Games Scale (BANGS) focuses on measuring autonomy, competence, and relatedness in play settings. While BANGS is grounded in SDT, our approach differs in its inductive derivation of factors based on actual player experiences [28].

Given the inductive nature of our scale development, no a-priori theoretical structure was imposed. However, the factors presented here suggest an intuitive overlap with major happiness frameworks. Table 1 summarizes these tentative correspondences. We emphasize that SDT, PERMA, and Flow are employed here solely as theories for interpretation; they did not guide item generation.

2.5. Problem identification

The above discussion highlights the importance of developing tools to measure happiness in games. Therefore, the purpose of this study is to provide an empirically validated multidimensional assessment tool for measuring happiness in games, “Game-Induced Happiness Scale (GIHS)” and to clarify how these dimensions, including specific game elements, are related to educational and psychological outcomes. The GIHS was developed inductively from player-generated responses, without reference to any specific pre-existing theoretical frameworks. Thus, interpretations of the extracted factors in this section are post-hoc

Table 1

Major happiness frameworks and representative game-play manifestations.

Framework/Construct	Core definition	Typical in-game manifestation
SDT–Autonomy [11]	Volitional choice	Sandbox exploration; narrative branching [26]
SDT–Competence [11]	Skill mastery	Adaptive difficulty, score-attack modes [11,26]
SDT–Relatedness [11]	Social bonding	Guild chat, co-op quests, voice comms [4]
PERMA–Positive Emotion [13]	Pleasant affect	“Cozy” games; aesthetic feedback loops [21]
PERMA–Engagement [13]	Deep absorption	Balanced challenge-skill (flow) [14,23]
PERMA–Relationships [13]	Supportive ties	Long-term multiplayer alliances [4]
PERMA–Meaning [13]	Purpose or meaning beyond self	Moral-choice narratives; world lore [29]
PERMA–Accomplishment [13]	Goal attainment	Trophies, 100% completion, leaderboards [20]
Flow–Challenge–Skill balance [14]	Skill, Challenge	Speed-run or rhythm-precision tasks [23]
Flow–Clear feedback [14]	Immediate response	Hit markers, combo meters, adaptive UI [23]

†These theoretical frameworks –Self-Determination Theory (SDT), PERMA, and Flow theory– were originally formulated within general psychological research. However, they offer valuable interpretive perspectives for analyzing happiness in the context of digital games. This table presents a post hoc theoretical integration of the inductively derived GIHS factors, illustrating how fundamental psychological constructs may be reflected in game environments through concrete design features and player experiences.

and not based on a-priori theoretical assumptions. This inductive approach contrasts with theory-driven instruments such as BANGS [28], which is grounded explicitly in SDT and designed to measure need satisfaction along predefined theoretical dimensions. In contrast, the GIHS aims to capture a broader and potentially more ecologically valid spectrum of game-induced happiness factors that emerge directly from players’ subjective experiences. After scale development, we examine how the resulting factors may align with established models (SDT, PERMA, Flow), but emphasize that these perspectives did not constrain item generation or initial factor derivation.

The remainder of this paper is structured as follows: Section 3 explains the analytical methods, Section 4 presents the results, Section 5 provides the discussion, and Section 6 provides the conclusion.

3. Methods

3.1. Survey participants and survey procedure

We conducted an online survey using Rakuten Insight in December 2023. The respondents comprised 500 Japanese university and graduate students (225 males and 275 females) with a mean age of 21.68 years (SD = 1.93). The valid response rate was 100.0%, and the average survey completion time was approximately 20 min. Only students who regularly played digital games participated in the survey, and their average daily game-playing time on weekdays was used as a screening criterion. The mean daily game-playing time was 2.31 h (SD = 2.69). Participant attributes are shown in Table 2. As shown in Table 2, when comparing males and females, males had higher daily gaming time, spending amount of in-game (monthly in-game spending), and game purchase amount (monthly software spending).

To ensure participants’ confidentiality, no personally identifiable information, such as names, student IDs, email addresses, or university names, was included in the survey questions. Consent to participate was obtained at the time of the survey. Respondents were compensated with reward points by the survey company. This study was approved by the ethics committee of Tokushima University, where the first author was affiliated (Approval No.01, 2022).

Table 2

Descriptive statistics for daily average play time, monthly in-game spending and monthly game software spending (male, female, total).

	Daily gaming time Mean $\pm$ SD (hours)	Monthly in-game spending Median [Q ₁ –Q ₃] (Japanese Yen)	Monthly game software spending Median [Q ₁ –Q ₃] (Japanese Yen)
Male	2.89 $\pm$ 3.21	0 [0–1000]	250 [0–2000]
Female	1.83 $\pm$ 2.07	0 [0–0]	0 [0–500]
Total	2.31 $\pm$ 2.69	0 [0–500]	0 [0–1000]

3.2. Survey items

This study pooled candidate items in a preliminary survey, created survey items from those items, conducted the main survey, and developed a scale. Notably, no a-priori reference to SDT, PERMA, Flow, or other happiness frameworks was made in this study. The survey items were developed through a preliminary study aimed at creating a scale to measure happiness in digital gaming. In August 2023, an open-ended questionnaire was conducted with 114 university students specializing in digital game development, focusing on happiness related to digital gaming. This group was chosen because students with development experience can articulate gameplay elements and emotional responses more precisely than general players, providing richer qualitative data for item generation.

The preliminary items were further refined based on feedback from university teachers specializing in educational psychology and faculty members in departments of game development at Japanese universities. Using cluster analysis with KH Coder 3, the initial 285 open-ended responses were grouped into 19 thematic categories. To improve clarity and relevance, a content validation procedure was conducted involving experts in game design and educational psychology, who independently evaluated each item on a 4-point scale for clarity and relevance (Content Validity Ratio method, CVR [30]). Items with a CVR value below 0.78 were excluded prior to the final selection.

The survey question was framed as follows: “During digital gaming (including video games and smartphone games) or related activities, do you feel ‘happiness’, ‘joy’, or ‘enjoyment’ when the following elements are present?” Responses were measured on a 5-point Likert scale ranging from 5: “Strongly agree” to 1: “Strongly disagree”.

To assess the criterion-related validity of the happiness scale for gaming, the subjective well-being scale outlined in Table 3 [31] was used. Respondents evaluated items on a 4-point Likert scale ranging from 4: “Strongly agree” to 1: “Strongly disagree”. The total score of these items was calculated to determine the respondents’ overall happiness. The subjective well-being scale utilized in this study has established reliability and validity in Japanese contexts, and its use was deemed appropriate for this research.

3.3. Analysis procedure

A preliminary analysis was conducted for each of the 143 items derived from the preliminary study to determine whether factor analysis could be performed on the data. Exploratory factor analysis (EFA) was then conducted using maximum likelihood estimation and promax rotation to develop the digital game happiness scale. SMC parallel analysis was used to determine the initial number of factors. Theoretical frameworks were consulted only after factor extraction for interpretive purposes. To evaluate the factors, confirmatory factor analysis (CFA), as well as assessments of criterion-related validity, content validity, convergent validity, and discriminant validity, were performed. All analyses were conducted using R version 4.3.2. All statistical results were rounded to two decimal places unless greater precision was needed for interpretation.

Table 3
Japanese version of the subjective well-being measurement scale [31].

No.	Question
1	Do you find life interesting?
2	Compared to the past, how is your life at present?
3	Overall, how happy do you feel about what you have done in the past few years?
4	When things don't go the way you want them to, do you think you can handle the situation adequately?
5	When you encounter a crisis (something that could destroy your life), do you feel confident that you will have the courage to face and solve it?
6	Do you feel confident that you can cope with what will happen in the future if you continue with your current course of action?
7	Do you think you have achieved the expected standard of living and social status?
8	How successful or unsuccessful do you feel you have been in your career?
9	Have you accomplished what you set out to do?
10 ^a	Do you feel that your life is boring or uninteresting?
11 ^a	Do you worry about your future?
12 ^a	Do you feel that your life is meaningless?
13	Do you sometimes feel a sense of belonging, of being an integral and indispensable part of your environment?
14	Do you have moments of extreme happiness?
15	Do you sometimes feel happy that you are part of the great family of mankind?

^a Invert Scale.

4. Results

4.1. Preliminary analysis

As part of the preliminary analysis, the data were checked for ceiling and floor effects, neither of which were observed. This finding suggests that the items on the digital game happiness scale provide an appropriate level of difficulty and a wide range of response variability. Next, the Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy was assessed, confirming that all items on the scale had a KMO index of 0.94 or higher. According to the criteria, this result is rated as “excellent” [32]. Generally, a KMO index of 0.90 or above indicates that the data are highly suitable for factor analysis [32,33]. Based on these findings, the data in this study were deemed highly appropriate for factor analysis. The KMO index measures the proportion of variance among variables that might be common variance, assessing the suitability of a correlation matrix for factor analysis [34]. In this study, the high KMO index values, along with the result of ceiling and floor effects, suggest that the digital game happiness scale is well-suited for a wide range of player populations.

4.2. Development of the measurement scale

Exploratory factor analysis (EFA) was conducted using maximum likelihood estimation and Promax rotation, with an initial solution based on the eight factors suggested by parallel analysis and diagonal SMC parallel analysis. The factor loading cutoff was set at 0.50, though it should be noted that there is no universally consistent standard for this threshold [33,34]. Typically, depending on sample size and research context, a cutoff range between 0.30 and 0.60 is considered appropriate [35]. In this study, items with factor loadings below 0.50, as well as items with loadings of 0.50 or higher on multiple factors, were removed to ensure interpretive clarity [36].

This iterative process led to a 5-factor structure being identified as the most appropriate, considering interpretability, theoretical validity, and statistical fit. This decision aligns with the emphasis on “interpretability”, a widely used criterion for determining the number of factors to extract in exploratory factor analysis [37]. Additionally, the

removed items were carefully reviewed in terms of their factor loadings and correlations across multiple factors to confirm that the resulting factor structure was theoretically valid.

This approach would strengthen the reliability and validity of the factor structure identified through EFA. The results are summarized in Table 4.

Factor I includes elements such as interaction with players from different generations and cultures, collaboration within teams or guilds, and communication via social networks. These factors suggest that in-game social interactions and bonding are strengthened. They encourage players to form social connections and participate in communities through shared experiences in the game. Therefore, Factor I was named “Social Interaction and Communication”.

Factor II focuses on strategic thinking and skill development within games. It encompasses abilities such as dodging enemy attacks, executing efficient strategies, and achieving challenging endings, reflecting the refinement of strategic thinking and the enhancement of technical skills. Hence, this factor was labeled “Strategic Gameplay and Skill Enhancement”.

Factor III highlights the unique traits and personalities of game characters. It includes elements such as beautiful character illustrations, captivating voiceovers, and unique character settings, which encourage players to empathize with game characters and immerse themselves more deeply in the game. Consequently, we named this factor “Character Appeal and Personality”.

Factor IV captures elements that keep players engaged in games for extended periods. These include unique gaming experiences, worlds that cannot be experienced in reality, and long-term gameplay that promotes deep immersion in the game world. Thus, this factor was named “Long-Term Immersion and Engagement”.

Finally, Factor V focuses on economic incentives and rewards within the game. This includes earning virtual currency, achieving economic success in the game, and obtaining significant rewards, representing the process through which players achieve economic goals and rewards. Therefore, Factor V was named “In-Game Economy and Rewards”.

4.3. Examination of the reliability and validity of the scale

4.3.1. Descriptive statistics of scale scores

Descriptive statistics for each factor are presented in Table 5. Among them, Factor V tends to have a relatively lower value. Means ranged from 2.87 to 3.60 on a five-point scale, and standard deviations from 0.85 to 1.09, indicating adequate response dispersion and the absence of ceiling or floor effects. Absolute skewness values were all below 0.45 and kurtosis never exceeded 0.24, well within the guidelines for univariate normality ($|\text{skew}| < 2$, $|\text{kurtosis}| < 7$ [38]).

4.3.2. Confirmatory factor analysis

To further confirm the stability of factor loadings and model fit, confirmatory factor analysis (CFA) was conducted.

Based on the results of exploratory factor analysis, a model was developed in which the items corresponding to each factor were influenced by five hypothesized latent factors. The validity of this model was tested using CFA. The model assumed covariance among all latent factors and no correlation among error terms. The CFA results yielded the following fit indices: Comparative Fit Index (CFI) = 0.886, Tucker-Lewis Index (TLI) = 0.878, Root Mean Square Error of Approximation (RMSEA) = 0.065, and Standardized Root Mean Square Residual (SRMR) = 0.053. While the model's CFI and TLI slightly fell below the threshold, they still indicated an acceptable model fit. In particular, RMSEA = 0.07 fell within the acceptable range (< 0.08), and SRMR = 0.05 satisfied the criteria for good fit (< 0.08). Although CFI and TLI were slightly below the rigorous cutoff of 0.90, the absolute fit indices (RMSEA = 0.065, SRMR = 0.053) were well within the recommended ranges, suggesting that the model adequately captures the underlying structure given the complexity of the 41-item scale [39].

Table 4
Results of the exploratory factor analysis ($N = 500$).

Item	I	II	III	IV	V
Factor I: Social Interaction and Communication					
1. Interact with a variety of players	0.92	−0.01	−0.05	−0.02	−0.07
2. Communicate through in-game chat	0.88	−0.04	−0.07	−0.04	0.03
3. Interact with different generations	0.85	−0.06	0.01	−0.07	−0.02
4. Exchange various in-game information	0.81	−0.03	−0.05	0.08	−0.03
5. Communicate through social networking sites	0.77	−0.22	0.17	−0.05	0.04
6. Connect with others in-game	0.74	−0.15	0.10	0.03	0.04
7. Play with users from overseas	0.69	−0.03	0.03	−0.07	0.00
8. Communicate in teams or guilds	0.66	0.20	−0.05	−0.08	0.11
9. Share goals within a team or guild	0.65	0.17	−0.05	0.05	0.01
10. Plan strategies to achieve common goals	0.63	0.18	−0.01	0.01	0.03
11. Collaborate with peers to solve puzzles	0.55	0.19	−0.03	0.17	−0.09
12. Work with peers on difficult challenges	0.55	0.19	−0.01	0.12	−0.03
Factor II: Strategic Gameplay and Skill Enhancement					
13. Successfully evaded enemy attacks	−0.01	0.86	−0.11	0.13	−0.11
14. Defeated multiple opponents one-by-one	0.09	0.81	−0.08	0.05	−0.05
15. Defeated an enemy with a counter move	0.01	0.79	0.04	−0.06	−0.00
16. Defeated an enemy without damage	−0.07	0.78	−0.06	−0.05	0.10
17. Defeated an enemy by considering the path	0.04	0.76	0.01	0.06	−0.05
18. Defeated an enemy with highest efficiency	0.01	0.74	0.01	0.01	−0.04
19. Got a good ending	−0.04	0.66	0.18	0.06	−0.04
20. Saw the true ending	−0.04	0.56	0.14	0.01	0.13
21. Saw all endings	−0.10	0.55	0.25	−0.09	0.17
Factor III: Character Appeal and Personality					
22. Beautiful illustrations of characters	−0.05	−0.01	0.80	0.10	−0.04
23. Good character voices	0.02	0.00	0.80	−0.01	−0.02
24. Character has personality	−0.01	−0.01	0.79	0.14	−0.15
25. Character is cool/cute	−0.05	0.04	0.76	0.13	−0.09
26. Character has a backstory	−0.01	0.02	0.76	−0.10	0.05
27. Character's setting is deep	−0.07	−0.03	0.74	0.14	0.04
28. Many conversations between characters	0.05	0.04	0.70	−0.13	0.08
29. Obtained assets of favorite character	0.15	−0.05	0.66	0.08	−0.02
Factor IV: Long-Term Immersion and Engagement					
30. Playable for many years	−0.08	0.06	−0.03	0.77	−0.06
31. Replayable after clearing	−0.05	0.09	−0.06	0.77	−0.03
32. Experiences unique to games	0.06	−0.03	0.07	0.71	0.05
33. Worldview beyond real life	0.07	−0.10	0.10	0.71	0.11
34. Games can be started for free	−0.01	−0.04	−0.06	0.67	0.01
35. Easy to start the game	−0.01	0.04	−0.04	0.62	−0.00
36. Immersed in the game world	0.01	0.04	0.21	0.62	−0.01
37. Characters are detailed and well-developed	−0.06	−0.02	0.33	0.55	0.04
38. Learnt new things through the game	0.03	0.10	0.09	0.55	0.11
Factor V: In-Game Economy and Rewards					
39. Obtained cash-convertible virtual currency	−0.01	−0.09	−0.08	0.07	0.88
40. Able to earn real money in the game	−0.01	−0.02	−0.07	0.02	0.87
41. Made a lot of money in the game	−0.00	0.07	−0.02	−0.01	0.70
<i>SS loadings</i>	6.56	5.11	4.91	4.24	2.21
<i>Proportion variance</i>	0.16	0.13	0.12	0.10	0.05
<i>Cumulative variance</i>	0.16	0.29	0.40	0.51	0.56
I	1.00	0.56	0.50	0.49	0.32
II		1.00	0.63	0.66	0.47
III			1.00	0.70	0.34
IV				1.00	0.31
V					1.00

Therefore, these results suggest that the model reasonably represents the data. CFA indicated acceptable – though suboptimal – fit. Although the model did not show a good fit due to CFI/TLI being below 0.90, considering the complexity of the scale, the results support acceptable model validity.

4.3.3. Internal consistency reliability

Table 6 presents Cronbach's α and McDonald's total ω_t for the five latent factors. Internal-consistency coefficients were uniformly high, with α ranging from 0.84 to 0.94 and the parallel ω_t values showing the same pattern (0.84–0.94). All estimates exceed the conventional acceptability benchmark of 0.70 and are well above the more stringent 0.80 threshold recommended for applied measurement contexts. Hence, each item set demonstrates strong internal homogeneity, supporting the use of factor scores in subsequent structural analyses. Although,

factor V showed the lowest coefficient ($\alpha = 0.84$, $\omega_t = 0.84$), but these values were also within the acceptable range, indicating that these reward-related items form a reliable subscale despite their conceptual uniqueness.

4.3.4. Convergent validity

Convergent validity was evaluated using composite reliability (CR) and average variance extracted (AVE). As shown in Table 7, all factors met the recommended thresholds of $CR \geq 0.70$ and $AVE \geq 0.50$, except for factor I ($AVE = 0.45$), which did not meet the criteria. However, since all standardized loadings exceeded 0.50 and were theoretically consistent in content, it is considered appropriate to retain factor I.

Table 5
Factor-level descriptive statistics ($N = 500$).

Factor	Mean	SD	Skew	Kurtosis
I: Social Interaction and Communication	3.20	0.91	−0.43	−0.14
II: Strategic Gameplay and Skill Enhancement	3.34	0.94	−0.41	0.05
III: Character Appeal and Personality	3.46	0.95	−0.41	−0.12
IV: Long-Term Immersion and Engagement	3.60	0.85	−0.33	−0.21
V: In-Game Economy and Rewards	2.87	1.09	−0.06	−0.67

Table 6
Internal-consistency coefficients for the five factors ($N = 500$).

Factor	Cronbach α	McDonald ω_i
I: Social Interaction and Communication	0.94	0.94
II: Strategic Gameplay and Skill Enhancement	0.92	0.93
III: Character Appeal and Personality	0.92	0.92
IV: Long-Term Immersion and Engagement	0.91	0.91
V: In-Game Economy and Rewards	0.84	0.84

Table 7
Composite Reliability (CR) and Average Variance Extracted (AVE).

Factor	AVE	CR
I: Social Interaction and Communication	0.45	0.88
II: Strategic Gameplay and Skill Enhancement	0.54	0.93
III: Character Appeal and Personality	0.53	0.91
IV: Long-Term Immersion and Engagement	0.57	0.91
V: In-Game Economy and Rewards	0.68	0.86

Table 8
HTMT (below diagonal) and 95% bootstrap CI (above diagonal) for 5-factor model ($R = 5,000$).

Factor	I	II	III	IV	V
I	–	[0.50, 0.68]	[0.44, 0.63]	[0.43, 0.63]	[0.25, 0.46]
II	0.60	–	[0.60, 0.75]	[0.63, 0.78]	[0.44, 0.62]
III	0.51	0.67	–	[0.69, 0.82]	[0.29, 0.49]
IV	0.53	0.72	0.78	–	[0.25, 0.46]
V	0.34	0.51	0.37	0.36	–

Note: I: Social Interaction and Communication, II: Strategic Gameplay and Skill Enhancement, III: Character Appeal and Personality, IV: Long-Term Immersion and Engagement, V: In-Game Economy and Rewards.

4.3.5. Discriminant validity

To assess discriminant validity, we calculated the heterotrait-monotrait (HTMT) ratio. We decided that the HTMT criterion should be less than 0.85 and the 95% CI should not be more than 1.

Table 8 shows the heterotrait-monotrait (HTMT) ratio and its 95% percentile bootstrap confidence intervals (5,000 resamples). All ten HTMT estimates ranged from 0.34 to 0.78; 95% percentile bootstrap CIs ($R = 5,000$) did not include 1.00 (upper bounds = 0.46–0.82), supporting discriminant validity.

Furthermore, none of the 95% bootstrap confidence intervals approached 1.00 (upper bounds = 0.46–0.82), confirming that the latent factors are empirically distinct. The highest associations were observed between Factor II (Strategic Gameplay and Skill Enhancement) and Factor IV (Long-Term Immersion and Engagement) (HTMT = 0.72, CI = 0.63–0.78), and between Factor III (Character Appeal and Personality) and Factor IV (HTMT = 0.78, CI = 0.69–0.82). These strong yet sub-threshold correlations reflect that mastery and character engagement naturally accompany immersive gameplay experiences. In contrast, Factor V (In-game Economy and Rewards) showed the lowest associations with all other dimensions (HTMT = 0.34–0.51, CIs $\leq$ 0.62),

Table 9
Criterion-related validity: group differences in subjective well-being.

Factor	t	Sign.
I: Social Interaction and Communication	3.95	**
II: Strategic Gameplay and Skill Enhancement	3.28	**
III: Character Appeal and Personality	2.11	*
IV: Long-Term Immersion and Engagement	3.50	**
V: In-Game Economy and Rewards	1.41	<i>n.s.</i>

underscoring its conceptual independence. Considering that reliability ($\alpha = 0.84$ –0.94) and convergent validity (AVE = 0.53–0.68) were also satisfactory, the HTMT findings provide strong evidence that the 5-factor structure demonstrates adequate discriminant validity.

4.3.6. Criterion-related validity

To establish criterion-related validity, we formed upper and lower groups for each factor (median split, $N = 500$) and compared subjective well-being using independent t -tests (Table 9). Significant mean differences were observed for factors I–IV ($t = 2.11$ –3.95, $p < 0.05$), while no significant differences were observed for factor V ($t = 1.41$, *n.s.*). Although Factor V does not show a direct association with subjective well-being, it is considered to capture the hedonic aspects of digital play based on rewards [40,41].

4.3.7. Competing-model comparison

Based on the above results, since the criterion-related validity of Factor V (In-Game Economy and Rewards) was not significant, it was necessary to examine whether the inclusion of Factor V was appropriate. To this end, we compared the proposed five-factor model with an alternative four-factor model excluding Factor V, in which all reward-related items were loaded on the immersion factor. The five-factor solution demonstrated equivalent or slightly superior fit (Table 10); $\Delta CFI = +0.002$, $\Delta TLI = +0.001$, $\Delta RMSEA = -0.004$, and $\Delta SRMR = -0.001$, all well within the negligible-loss range [42]. Because the alternative four-factor specification was not strictly nested within the five-factor model, information criteria were also compared. As shown in Table 10, the four-factor model yielded lower AIC and BIC values (AIC = 47,871 vs. 52,109; BIC = 48,377 vs. 52,669).

While the information criteria (AIC/BIC) provided support for the more parsimonious four-factor model, it failed to incorporate the unique concept of “in-game economy and rewards” and did not maintain the conceptual granularity of the reward dimension, which is increasingly important in contemporary game design. Given that the objective of this study is to comprehensively evaluate the multifaceted nature of game-induced well-being, interpretability and theoretical breadth were prioritized over statistical parsimony, and the five-factor structure was retained.

4.3.8. Reliability of the scale

To examine the reliability of the developed scale, 35 students were asked to respond to the items associated with the identified factors. A follow-up survey was conducted two weeks later. The correlation coefficients between the variables obtained from the two surveys were calculated as test-retest reliability coefficients, yielding $r = 0.71$ ($p < 0.01$). This result suggests that the scale provides consistent measurement results over time and demonstrates relatively high reliability.

Additionally, as criteria for a test-retest validity, Nunnally and Bernstein [43] proposed that the test-retest reliability coefficient $r > 0.70$ indicates sufficient reliability in psychological research. Kline [44] also noted that values in the range of $r = 0.70$ to 0.80 are acceptable in many social science studies. The results of this study meet these criteria.

Table 10
Model-fit comparison between the 5- and 4-factor CFA ($N = 500$).

Model	χ^2	df	CFI	TLI	RMSEA	SRMR	AIC	BIC
5-factor (M_5)	2400	769	0.886	0.878	0.065	0.053	52,108.51	52,669.06
4-factor (M_4)	2219	659	0.884	0.877	0.069	0.054	47,871.27	48,377.03
Difference (M_5 - M_4)	181	110	+0.002	+0.001	-0.004	-0.001	-	-

5. Discussion

5.1. Reliability and validity of the measurement scale for happiness in digital gaming

This study aims to construct the GIHS based on actual user behavior. To achieve this, we utilized a mixed-methods pipeline (item generation, exploratory factor analysis (EFA), confirmatory factor analysis (CFA), and a battery of reliability/validity tests) to accumulate convergent evidence regarding the psychometric validity of the scale. Below, we present the results, discuss their implications, and integrate the findings to demonstrate the validity of the research.

Cronbach's α (0.84–0.94) and McDonald's ω_i mirrored one another, exceeding the 0.80 benchmark recommended for applied contexts (Table 6). Although the In-Game Economy & Rewards sub-scale (Factor V) produced the lowest coefficient, its value of .84 still represents “good” reliability [43]. Two-week test-retest reliability ($r = 0.71$) further corroborated temporal stability, aligning with the 0.70–0.80 range deemed acceptable for attitudinal measures [44]. Taken together, these indices demonstrate that both item homogeneity and score reproducibility meet contemporary psychometric standards.

EFA suggested a interpretable 5-factor solution that accounted for 56% of the common variance (Table 4). CFA yielded an acceptable – though not impeccable – fit (CFI = 0.886, TLI = 0.878, RMSEA = 0.065, SRMR = 0.053). While the model did not fully meet the strict ‘close fit’ criteria (e.g., CFI ≥ 0.95) recommended by Hu and Bentler [45], the RMSEA and SRMR values were within acceptable thresholds, supporting adequate model fit for behavioral data.

Convergent validity was affirmed by composite reliabilities (0.86–0.93) well above 0.70 and AVE values exceeding 0.50 for four of five factors (Table 7). Although the AVE for Factor I (0.45) was below the 0.50 threshold, composite reliability was high (CR = 0.88), and each loading exceeded 0.70. According to Fornell and Larcker [46], factors with AVE > 0.40 can be retained if CR is sufficiently high. Discriminant validity was supported via HTMT ratios (0.34–0.78) and 95% bootstrap CIs (upper bounds = 0.46–0.82) excluding 1.00 (Table 8), indicating that the five latent constructs are empirically distinct despite moderate inter-factor correlations.

The cumulative evidence positions the GIHS as a reliable, multi-faceted tool suitable for research and applied diagnostics. Each factor maps onto established well-being frameworks-social bonds (SDT relatedness), mastery (SDT competence/Flow), aesthetic engagement (PERMA engagement), immersion (Flow), and hedonic gain (short-term positive affect). Game developers can thus target specific GIHS dimensions to design for sustained happiness (e.g. balancing skill challenge to boost Factor II, enriching lore and narrative for Factor III). Educators and clinicians may likewise choose or adapt games that score highly on factors aligned with their objectives (e.g. Factor I for social skills training).

5.1.1. Confirmation of reasons for adopting factor V

To evaluate criterion-related validity, an independent samples t -test based on median splits of the upper and lower subject well-being scores was conducted, and the results showed that four out of five factors statistically significantly distinguished the high satisfaction group from the low satisfaction group (Table 9). The absence of significant differences in factor V suggests that reward-based enjoyment may be more closely associated with temporary pleasure than with global life

satisfaction. This aligns with the self-determination theory framework that distinguishes between extrinsic and intrinsic motivation [26].

Although Factor V did not correlate with life satisfaction, it may still contribute to players' emotional involvement during gameplay [47], consistent with findings on short-term affective engagement in learning contexts.

Moreover, as shown in Section 4.3.7, comparing models excluding Factor V with those including Factor V reveals that the 5-factor solution demonstrates better or equivalent fit, suggesting that the inclusion of Factor V is reasonable.

In summary, the inclusion of factor V is consistent with our goal of developing a comprehensive measure that captures diverse aspects of the gaming experience. While traditional subjective well-being measures play an important role in evaluating happiness, they may not fully capture the breadth of emotions and experiences elicited by digital games. In digital games, in-game economic systems and rewards provide players with a sense of achievement, control, and immersion-core elements of the gaming experience [26]. These emotions satisfy players' psychological needs, enhance the appeal of games, and contribute to players' well-being and satisfaction.

Therefore, the addition of a fifth factor to the scale is an important step toward comprehensively understanding the impact of digital games on players. This reflects the complexity of the gaming experience and the need for scales that can capture its diverse aspects.

5.2. Post-hoc theoretical integration

Building upon the empirical results in Section 4, this chapter integrates the findings and interprets their theoretical and practical implications. To enhance readability, the discussion first summarizes psychometric evidence before linking each factor to existing well-being theories.

5.2.1. Consideration by factor -relationship with SDT-

In the following, we will examine the content of the factors in detail using SDT as an example.

(1) Factor I: Social Interaction and Communication:

Factor I emphasizes that interactions with others during gameplay foster social connections and increase players' happiness. In particular, the experiences of empathy, friendship, and co-operation gained through online games and team play satisfy the “relatedness” needs of SDT. Cole and Griffiths [4] state that MMORPGs reinforce communication between players, reduce isolation, and support the idea that social bonds contribute to happiness. Furthermore, guilds and team features in online games serve as a foundation for sustained connections, enhancing players' social support and sense of belonging [20]. Therefore, Factor I is strongly associated with the “relatedness” element of SDT.

(2) Factor II: Strategic Gameplay and Skill Enhancement:

Factor II reflects the sense of achievement and growth gained from strategic play and skill improvement, and is closely related to the “competence” concept of SDT. Setting challenging yet achievable goals increases players' self-efficacy and happiness [20]. Features such as skill trees and ranking systems within games stimulate learning motivation and provide opportunities for growth [19]. This not only contributes to a sense of achievement but also has the potential to improve problem-solving and

cognitive skills in real-world contexts. Therefore, factor II is strongly related to the “competence” element of SDT.

(3) Factor III: Character Appeal and Personality:

Although not a direct element of the SDT framework, factor III (emphasizing the emotional impact of character appeal and personality) shows an indirect relationship with the elements of SDT. High freedom in character selection and customization enhances “autonomy”, while emotional connection and empathy toward characters strengthen “attachment”(like relatedness) [48]. Furthermore, immersive elements and settings not only deepen players’ immersion but also contribute to promoting happiness [14] suggesting that immersive narrative elements are expected to contribute to happiness. Therefore, Factor III is indirectly related to SDT elements, with its relevance becoming particularly strong in contexts where customization and narrative depth are prominent.

(4) Factor IV: Long-Term Immersion and Engagement:

Factor IV is an element related to long-term engagement and immersion in games, and is consistent with both “autonomy” and “relatedness” in SDT. Freely explorable story progression and open-world elements encourage players to act autonomously [11]. Deep immersion experiences function as psychological escape mechanisms, reducing stress and anxiety while improving well-being [19]. Therefore, Factor IV is related to the “autonomy” and “relatedness” elements of SDT.

(5) Factor V: In-Game Economy and Rewards:

Factor V represents the extent to which in-game economic incentives and rewards provide players with a sense of achievement and control, contributing to their happiness. When the reward acquisition process offers high freedom (“autonomy”) or allows players to demonstrate their skills or abilities (“competence”), Factor V aligns with the SDT framework. However, its relationship with “relatedness” may be limited, except when rewards emphasize social connections (e.g., cooperative play or socially motivated rewards). Additionally, caution is needed to avoid reward systems that may lead to excessive dependence or gambling-like behavior. While loot boxes and gacha mechanics may provide temporary excitement, they also carry risks of problematic gambling and excessive spending if not implemented responsibly [40,41]. In fact, freemium and “pay-to-win” monetization models may lead to excessive consumption patterns similar to gambling, emphasizing the importance of ethical design and social responsibility measures. Long-term happiness requires balance, ensuring that both paid and free users feel satisfied and are not placed at a disadvantage solely based on monetary investment. Properly balancing and capping in-game rewards can help avoid players accumulating rewards beyond enjoyment (an ‘overearning’ phenomenon identified by Hsee et al. [49]), keeping the reward experience fulfilling rather than excessive.

Therefore, factor V aligns with the “autonomy” and “competence” elements of SDT and is particularly effective when the reward acquisition process emphasizes these elements in a transparent and player-friendly manner.

These results indicate that the three basic psychological needs of SDT – autonomy, competence, and relatedness – contribute significantly to increasing happiness during gameplay. Game design that takes these factors into account has the potential to improve players’ psychological well-being while providing educational and entertainment value. Future research should incorporate complementary perspectives from SDT to investigate the multifaceted effects of digital games on well-being. Furthermore, factors beyond the SDT framework, such as the appeal and individuality of characters, play a role in improving the quality of the gaming experience and player satisfaction. Understanding these diverse aspects is essential for grasping the multifaceted dimensions of happiness in games.

5.2.2. Relationship between existing happiness and the GIHS

In this section, we attempt to interpret the happiness obtained from five games using three well-known happiness theories (SDT, PERMA, and Flow). Looking back on the flow of this study, we developed a bottom-up scale by extracting factors through exploratory factor analysis and utilized theory as a framework for interpretation. Table 11 summarizes how the extracted factors can be understood retrospectively from the perspectives of SDT, PERMA, and Flow theory. Since the theoretical models were not explicitly applied during item generation, this table serves as a bridge for interpretation. Table 11 shows that the five factors identified in this study are related to existing happiness theories. As shown in Table 11 offers a post-hoc interpretive bridge mapping the emergent factors onto SDT, PERMA, and Flow; this mapping is interpretive rather than predictive and was not used during item generation. Besides, the factors are (i) SDT’s need satisfaction, (ii) PERMA’s multifaceted happiness, and (iii) Flow theory’s optimal experience, with game-specific elements (factors III and V) complementarily incorporated. This suggests that the scale explicitly demonstrates a structure that mediates existing theories through so-called “theoretical mediation” and ensures external validity. By explicitly demonstrating a structure that ensures external validity through theoretical mediation rather than relying solely on inductive extraction, this scale enhances its theoretical validity. Furthermore, although this scale is inductively derived, the content shown in Table 1 supports the external validity of the obtained scale and suggests a promising direction for theoretical construction that bridges everyday happiness and game-specific happiness.

While the BANGS scale provides a robust SDT-based framework [28], it does not capture hedonic or aesthetic facets such as character appeal or in-game economies. In contrast, the GIHS includes these experiential dimensions, allowing for a more holistic account of happiness in games beyond purely need-based perspectives.

5.3. Applications of this scale

The scale developed in this study has several potential applications, particularly in contributing to the improvement of players’ psychological well-being. Games have been shown to reduce stress, improve mood, and have positive effects on mental health. By using this scale to gain a deeper understanding of happiness in games, it will be possible to further explore these positive effects and maximize their benefits. For example, a practical application focusing on Factor I could be to enhance features such as in-game chat, voice chat, and cooperative tasks to promote positive social interaction among players and improve their sense of happiness. For Factor II, practical applications could include introducing features that visualize players’ skills and incorporating game designs that gradually challenge players with more difficult tasks. Such designs are expected to increase players’ sense of competence and further enhance their sense of happiness. For Factor III, practical examples include enhancing character customization options and developing storylines that capture players’ interest, which can increase player satisfaction. For Factor IV, open-world game design and the provision of flexible game environments can promote continuous engagement and well-being. Factor V includes activities that carry the risk of addiction. Therefore, it is important to design a balanced reward system that provides appropriate rewards for players’ efforts while avoiding excessive dependence on in-game purchases and enhancing happiness. Furthermore, utilizing happiness data in game design can create more appealing and satisfying game experiences, ultimately increasing user engagement and loyalty. In educational games, this scale can be used to visualize the positive impact of player happiness on learning outcomes and provide valuable insights for designing enjoyable learning experiences. Games that promote social connection and community building can also benefit from this scale’s insights to foster positive social interactions among players. The happiness measured by GIHS can also be used to assess the risk of excessive gameplay and addiction.

Table 11

Post-hoc interpretive mapping between the five emergent factors and three well-being frameworks: Self-determination theory, PERMA, Flow.

Factor	Interpretive SDT alignment	Interpretive PERMA alignment	Relevant Flow component
I: Social Interaction and Communication	Relatedness	Relationships (R)	Social presence/shared goals
II: Strategic Gameplay and Skill Enhancement	Competence	Long-term Achievement/Accomplishments (A)	Challenge-skill balance/clear feedback
III: Character Appeal and Personality	Autonomy ^a	Engagement (E)/Meaning (M)	Deep focus/transformation of time
IV: Long-Term Immersion and Engagement	Autonomy (Volitional Enjoyment)	Positive Emotion (P)	Effortless control/absorption
V: In-Game Economy and Rewards	Limited Alignment (Positive Emotion, Autonomy, Competence)	Short-term Achievement (A)	Immediate feedback/Reward loop

^a Narrative immersion often enhances a sense of player agency, loosely aligning with SDT-autonomy. Dashed cells indicate interpretive, not predictive, correspondence. Tick-marks or color-coding can be added during layout if desired.

This supports the development of strategies to promote healthy gaming habits and protect players from negative influences. Furthermore, this scale can identify factors that maximize player happiness, support the design of more satisfying games, and provide guidelines for measuring the effectiveness of educational games and enhancing their educational value. It also plays a role in reducing stress within games and evaluating factors that produce positive psychological effects, thereby promoting healthy game use.

5.3.1. Practical implications

The validated 5-factor game-induced happiness scale provides multiple practical implications for educators, game designers, and policymakers. For educators, this scale serves as a tool to diagnose and evaluate whether so-called “digital games” promote psychological and social outcomes for students. For example, by identifying games that score high on factors such as “in-game communication and sociality” or “strategic gameplay and skill improvement”, teachers can intentionally select or recommend titles that promote teamwork, problem-solving, and personal growth. Furthermore, by tracking changes in students’ happiness before and after game-based interventions, educators can measure the effectiveness of serious games aimed at learning and social development.

For game designers and developers, each element of GIHS directly contributes to improving the user experience. For example, improving cooperative play and communication tools to emphasize Factor I can increase player satisfaction and retention rates. Integrating meaningful challenges and visible skill improvement (Factor II) supports long-term engagement and skill mastery. Well-developed characters and stories (Factor III) enhance emotional investment and immersion, while game features that promote sustained play (Factor IV) encourage healthy, long-term participation. Finally, by carefully balancing in-game rewards and economic systems while considering ethical considerations (Factor V), short-term enjoyment can be increased while minimizing the risk of problematic play.

For policymakers and mental health professionals, GIHS provides a foundation for transitioning from simple time-based regulations (such as screen time limits) to more nuanced, well-being-centered guidelines. This metric can help identify high-risk groups, provide public health advice, and evaluate the social value of game interventions. By monitoring happiness profiles according to players’ demographic attributes, interventions can be optimized to maximize positive outcomes and mitigate potential harms, supporting the development of healthy gaming policies and prevention programs for gaming-related issues such as gaming addiction.

5.3.2. Theoretical implications

This study provides multiple theoretical contributions to the fields of happiness research and digital game research. First, the inductive, bottom-up development of the GIHS provides empirical evidence of the multifaceted nature of happiness derived from digital games and suggests that happiness from games may blur the distinction between pleasure-based and meaning-based happiness. The identification of five independent yet interrelated factors-social interaction, strategic skill development, character appeal, immersion, and in-game rewards-shows that happiness in games is supported by both game-specific processes and general psychological processes. Second, GIHS is related to established happiness frameworks such as SDT, PERMA, and Flow theory, enriching existing theories. GIHS maps onto key concepts in these theories. Furthermore, factors that are not adequately addressed in classical models (e.g., character appeal) highlight the importance of immersion in game-specific narratives and aesthetics. Third, by providing psychometrically robust indicators rooted in players’ experiences, this study advances academic discussions on game-based well-being and lays the groundwork for comparative research across cultures, genres, and player groups. A data-driven, theory-informed approach advocates integrating the player’s perspective into well-being theory and highlights the potential of games as complex, multifaceted activities that foster a wide range of psychological benefits.

6. Conclusion

6.1. Summary of the study

The purpose of this study was to develop Game-Induced Happiness Scale (GIHS), a measure of happiness resulting from digital game play, and to evaluate its reliability and validity. A 41-item scale for measuring happiness in digital games was developed, with a factor structure consisting of the following 5 main factors: communication and socialization within the game, strategic gameplay and skill development, character appeal and individuality, long-term appeal and immersion, and in-game economy and rewards. The scale was found to be reliable and valid. Furthermore, we demonstrated that digital games have multifaceted effects on users’ happiness and clarified basic associations with existing happiness theories such as Self-Determination Theory (SDT), PERMA, and Flow. Additionally, achievement and social connectedness significantly enhanced happiness, while the effects of in-game economy and rewards were limited. This difference suggests that it may depend on the type of game and individual play styles. The developed scale is expected to provide basic insights for the appropriate use of games in education, improving players’ mental health, promoting social interaction through games, and developing treatments for game addiction and disorders. Furthermore, the importance of game

design that enhances well-being while reducing the risk of addiction is suggested. This study constitutes the first attempt to develop a validated happiness scale tailored to digital games, offering an original contribution to both gaming and well-being research.

6.2. Limitations of the study

As described above, this study provides many insights and has significant originality and novelty, but it also has many limitations.

While the current sample included both male and female university students, the generalizability across age, culture, and gaming habits remains untested. Future research should examine scale invariance across international and genre-specific cohorts. Furthermore, a more precise operationalization of ‘game-induced happiness’ is needed to clarify boundaries with related constructs such as subjective well-being or life satisfaction.

1. Limited sample: Since this study targeted Japanese college students, the sample was biased. Factors such as age and gender were not taken into account in the scale development. Future studies should target diverse demographic groups and conduct longitudinal studies to verify the applicability of the scale.
2. Gameplay parameters: Happiness in games may vary depending on parameters such as heavy users and casual users, game type, game platform, gender, and region. Future research should comprehensively investigate these factors and conduct international studies.
3. Psychological factors: The relationship between psychological factors related to happiness in digital games and gameplay motivation remains unclear. To advance educational applications, mental health improvement, social interaction promotion, and addiction treatment, it is important to investigate how these factors are related to happiness.
4. Definition of happiness: To deepen this study, it is necessary to refine the concept of “happiness in games”. Although there is abundant research on happiness, subjective well-being, and digital well-being, a definition of happiness in the context of games is essential.
5. Elucidation of factors that do not fit existing definitions of happiness: Factors contributing to happiness in games go beyond those indicated by SDT, such as autonomy, competence, and relatedness. These factors greatly influence the quality of gameplay and player satisfaction. Expanding research to include factors that go beyond existing frameworks of happiness is necessary to comprehensively understand the factors that contribute to happiness.

Additional areas of focus include differences in happiness factors across game genres, causal relationships between game elements and happiness, and longitudinal studies tracking changes in happiness over time. Collaboration with game designers and developers has the potential to promote the practical application of these findings. Conducting such research will lead to a deeper understanding of happiness in digital games and contribute to improving players’ mental health, promoting appropriate gameplay, and addressing game addiction. These efforts will also enhance the social value of games. Furthermore, future studies should consider direct comparisons between the GIHS and existing instruments such as BANGS [28] to evaluate convergent and discriminant validity. Such comparisons would clarify the degree to which inductively and deductively derived measures capture overlapping or distinct aspects of game-related well-being. Moreover, future research is expected to build on the results of studies on the relationship between happiness and motivation in games [50], and the relationship between loot boxes and motivation in games [51], leading to further advanced research.

CRediT authorship contribution statement

Masanori Fukui: Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Wanda Nugroho Yanuarta:** Writing – review & editing, Formal analysis, Validation, Resources. **Subuh Anggoro:** Writing – review & editing, Validation, Resources. **Eng Tek Ong:** Writing – review & editing, Supervision, Resources. **Khar Thoe Ng:** Writing – review & editing, Validation, Supervision.

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Declaration of competing interest

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Data availability

The authors do not have permission to share data.

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